ZAP.COOKING · ENGINEERING

Mixed replies, one thread.

Nostr answers a note two ways: **kind 1 replies** (NIP-10) and **kind 1111 comments** (NIP-22). Here is how we read, write and count both — in one tree.

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01 — THE FAILURE

A working query hides half the thread

A comment carries its `e` = the note it answers, and its `E` = the thread root. So `#e = root` — the query that works for kind-1 threads — only ever finds **top-level comments**. Every reply-to-a-comment below them is invisible.

NOTE · kind 1
the root note

REPLY · kind 1
a NIP-10 reply

```
e [root, "", "root"]   e [note, "", "reply"]
```

root + reply **markers**; every descendant repeats the root

COMMENT · kind 1111
replying to that note

```
E [root]   K [1]   P [root-author]   ← root scope
e [note]   k [1]   p [note-author]   ← parent scope
```

COMMENT · kind 1111
replying to that comment

```
E [root]   — unchanged, — still the root
e [comment-above]   k [1111]
```

uppercase E always names the root; lowercase e walks parent to parent.
Slot 3 of a comment's e is the parent **author** — never a NIP-10 marker.

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02 — THE READ

One REQ, two sibling filters

```
["REQ", "thread",
  { "kinds": [1, 5, 1111], "#e": [root, focal] },
  { "kinds": [1111],       "#E": [root, focal] } ]
```

The `#e` filter walks the NIP-10 tree (plus deletions); the `#E` filter returns the whole comment tree. Relays `OR` sibling filters — merged into one object, the tags `AND` and match nothing new. Verified live: on a four-deep chain, `#e` returns 1 event, `#E` returns 4.

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03 — THE WRITE

Reply in the dialect the thread speaks

`target = comment` → **kind 1111**
copy its `E/A/I + K + P` root scope verbatim, append `e / k / p` pointing at the parent

`target = anything else` → **kind 1**
plain NIP-10 tags — the default, because every client renders them everywhere

`comment has no E/A/I` → **kind 1 fallback**
never publish a reply that nobody — including us — could place in a tree

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04 — THE PARITY

A comment is a reply

COUNTS
toward the parent it answered — never the root

NOTIFICATIONS
"replied to your comment" · "your post" · "in your thread"

DETAILS
`KIND 1111 · COMMENT` beside "Posted via…"

⚡ **zap.cooking**

Ported from **zapcooking_ios** PRs **#161** + **#145** · conventions confirmed against Ditto's client code

`#E` indexing verified on relay.azzamo.net · nostr.wine · nos.lol · relay.ditto.pub